APM5557: Partial Differential Equations
Functions, Equivalence Classes, and Distributions

The three kinds of object are related by a chain of maps:

$$\text{ordinary functions} \longrightarrow \text{equivalence classes} \longrightarrow \text{distributions.}$$

Each map forgets some information and makes more operations possible. In particular, every object in the chain becomes differentiable once you reach the distributions.

1 Ordinary functions

What they are. A rule $f : \Omega \to \mathbb{R}$ (or $\mathbb{C}$) with a definite value $f(x)$ at every point.

Equality. $f = g$ means $f(x) = g(x)$ for *every* x.

Typical spaces.

- $C(\overline{\Omega})$, with $\|f\|_\infty = \sup |f|$
- $C^k(\overline{\Omega})$, with $\|f\| = \sum_{|\alpha| \le k} \sup |\partial^\alpha f|$
- $C^{k,\alpha}$, the Hölder spaces, and Lipschitz functions
- $C^\infty(\Omega)$, and $C_c^\infty(\Omega)$, the smooth functions with compact support (the test functions)

Language. "f is continuous at x_0," "$f(x_0) = 3$," "$f'(x_0)$ exists." Evaluating at a point, $f \mapsto f(x_0)$, is a continuous linear functional on these spaces.

Derivatives. Classical, defined pointwise as limits of difference quotients, and they may fail to exist.

2 Equivalence classes (functions defined up to measure zero)

What they are. $[f] = \{g : g = f \text{ a.e.}\}$. Usually you just write f and call it a *representative*.

Why we need them. $\int |f|^p = 0$ implies only $f = 0$ a.e., not $f \equiv 0$. On functions, $\|\cdot\|_p$ is therefore just a seminorm. It becomes a true norm only after you identify functions that agree a.e. Formally,

$$L^p = \mathcal{L}^p / \{f : f = 0 \text{ a.e.}\}.$$

Equality. $f = g$ in L^p means $f = g$ almost everywhere.

Typical spaces.

- $L^p(\Omega)$ for $1 \le p \le \infty$; L^∞ uses the essential supremum

- $L^1_{\text{loc}}(\Omega)$, the functions integrable on every compact subset
- The Sobolev spaces $W^{k,p}(\Omega)$ and $H^k = W^{k,2}$, with $H^1_0 \subset H^1$

Language.

- "$f(x_0)$" is **meaningless**, because a single point has measure zero and you can change f there freely.
- "$u \in H^1(a,b)$ is continuous" means that u **has a continuous representative**, and you agree to use it. Since two continuous functions that agree a.e. agree everywhere, that representative is unique, which is why we can treat C as a subspace of L^p.
- Boundary values such as "$u = 0$ on $\partial\Omega$" in higher dimensions need **trace theory**, because $\partial\Omega$ has measure zero.

Derivatives. Weak derivatives. $g = u'$ means

$$\int u\,\varphi' = -\int g\,\varphi \quad \text{for every } \varphi \in C_c^\infty.$$

The result is again an equivalence class.

3 Distributions

What they are. Continuous linear functionals $T : C_c^\infty(\Omega) \to \mathbb{R}$. They aren't functions of x at all. They are defined only through how they act on test functions, written $\langle T, \varphi\rangle$.

Equality. $T = S$ means $\langle T, \varphi\rangle = \langle S, \varphi\rangle$ for every test function φ.

Typical spaces.

- $\mathcal{D}'(\Omega)$, the dual of $C_c^\infty(\Omega)$
- $\mathcal{S}'(\mathbb{R}^n)$, the tempered distributions (the dual of the Schwartz space), which is where the Fourier transform lives
- $\mathcal{E}'(\Omega)$, the distributions with compact support
- Negative-order Sobolev spaces such as $H^{-1} = (H^1_0)'$, and H^s for $s < 0$

Spaces like $\mathcal{D}'$ and $\mathcal{S}'$ are **not normed**. They carry weak-* topologies: $T_n \to T$ means $\langle T_n, \varphi\rangle \to \langle T, \varphi\rangle$ for every φ. Spaces like H^{-1}, by contrast, are Hilbert spaces.

How functions fit in. Each $f \in L^1_{\text{loc}}$ defines a distribution by

$$\langle T_f, \varphi\rangle = \int f\,\varphi.$$

By the fundamental lemma of the calculus of variations, $T_f = T_g$ exactly when $f = g$ a.e. So it is the **equivalence classes**, not the individual functions, that embed injectively into $\mathcal{D}'$. A distribution of the form T_f is called *regular*. The δ function, p.v. $1/x$, and δ' are not regular.

Language.

- "$T \in \mathcal{D}'$ is in L^2" means $T = T_f$ for some $f \in L^2$.

- "Restricted to $(0,1)$, T is smooth" refers to its *singular support*, the set where T fails to be given by a smooth function.
- "The support of T" is the complement of the largest open set on which $T = 0$.

Derivatives. Every distribution has derivatives of every order, defined by

$$\langle T', \varphi \rangle = -\langle T, \varphi' \rangle.$$

The price is that you **can't multiply** two distributions in general. $\delta \cdot \delta$, for example, is undefined.

4 The chain of inclusions

$$C_c^\infty \subset C^k \subset C \hookrightarrow L^p_{\text{loc}} \subset L^1_{\text{loc}} \hookrightarrow \mathcal{D}'.$$

The arrow $C \hookrightarrow L^p_{\text{loc}}$ is injective on continuous functions. On arbitrary functions the map to classes is not injective, because many functions share one class. Sobolev spaces fit in the same way, for example

$$H^1_0 \subset L^2 \subset H^{-1} \subset \mathcal{D}',$$

which is the "Gelfand triple" used in weak formulations of PDEs.

5 Summary table

	Ordinary functions	**Equivalence classes**	**Distributions**
Object	value at every point	functions equal a.e.	functional on test functions
Equality	everywhere	almost everywhere	against every φ
$f(x_0)$ defined?	yes	no (only if a continuous representative exists)	no
Derivative	classical, may not exist	weak, may not exist	always exists
Products	yes	yes, if the integrability works out (e.g. $L^2 \cdot L^2 \subset L^1$)	not in general
Typical spaces	C, C^k, $C^{k,\alpha}$, C^∞	L^p, L^1_{loc}, $W^{k,p}$, H^k	$\mathcal{D}'$, $\mathcal{S}'$, $\mathcal{E}'$, H^{-s}
Structure	Banach (C^k), Fréchet (C^∞)	Banach / Hilbert	weak-* topology, generally not normed

6 Three examples where the levels give different answers

1. $f(x) = |x|$
   - *Classical:* differentiable everywhere except $x = 0$.
   - *a.e. / weak:* $f' = \operatorname{sign}(x)$, an L^∞ class. The missing value at 0 doesn't matter.
   - *Distributional:* $f'' = 2\delta$, which is not a function.
2. **Heaviside function** H
   - *Classical:* $H' = 0$ everywhere except 0.

- *a.e.:* the pointwise a.e. derivative is 0.
- *Weak:* H has **no** weak derivative in L^1_{loc}, so $H \notin W^{1,1}_{\text{loc}}$.
- *Distributional:* $H' = \delta$.

So "differentiable a.e." is **not** the same as "weakly differentiable."

3. **Cantor function** c

   - *Classical / a.e.:* c is continuous, and $c' = 0$ a.e.
   - *Distributional:* c' is the Cantor measure. That measure is singular, so it isn't an L^1_{loc} function, and $c \notin W^{1,1}$.

   This is why Sobolev functions in 1D must be *absolutely* continuous: continuity alone is not enough.